

Olist Brazilian E-Commerce

What Drives Customer Satisfaction?

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Ask

Business Task

Analyze the Olist Brazilian e-commerce dataset to identify which factors drive customer satisfaction (review scores), helping the company improve the customer experience and reduce negative reviews.

Data Source

Brazilian E-Commerce Public Dataset by Olist (Kaggle). ~100,000 orders from 2016-2018, split across 9 relational tables.

Tools

Python (pandas) in Google Colab for data cleaning and analysis; Tableau for visualization.

Prepare & Process (Data Cleaning)

Reviews table

- 99,224 rows, 7 columns. Key column is review_score (1-5).
- Key columns (review_score, order_id) have zero missing values.
- review_comment_title is 88% empty and review_comment_message 59% empty, because many customers rate without writing a comment (expected, not an error). No duplicate rows.

Decision: kept all rows (the score is what matters) and filled empty comment fields with "No comment" rather than dropping data.

Orders table

- 99,441 rows, 8 columns. Contains key delivery dates (estimated vs actual delivery to customer).
- 2,965 orders had no delivery date because they were never delivered (canceled, in transit, etc.), not a data error.
- Order status breakdown: 96,478 delivered, ~2,963 in other states.

Decision: kept only 'delivered' orders (96,478), since only those have a real delivery time to analyze. Also removed 8 orders marked "delivered" but missing a delivery date (data inconsistency). Final clean orders table: 96,470 rows.

Feature engineering

- Converted date columns from text to datetime format so delivery times could be calculated.
- delivery_days: total days from purchase to delivery.
- delivery_vs_estimated: days early (negative) or late (positive) versus the promised delivery date.
- Average delivery time: 12 days (median 10). On average Olist delivers ~12 days earlier than estimated (conservative estimates). Outliers exist: longest delivery 209 days; worst delay 188 days late.

Merging tables

- Merged the cleaned orders table with reviews using order_id (inner join), keeping only orders with both delivery data and a review score.

- Order items table (112,650 rows) has one row per product; grouped by order_id to get one row per order, summing product prices and shipping costs.
- Chained products + translation + items to link each order to its product category in English.

Analyze

Finding 1 — Delivery time is the strongest driver of satisfaction. Average delivery time drops steadily as scores rise: 20.8 days (1 star) down to 10.2 days (5 stars). 1-star customers waited roughly twice as long as 5-star customers. 5-star orders also arrived ~13 days ahead of the promised date, vs ~4 days early for 1-star orders — beating the delivery estimate matters, not just speed.

Finding 2 — Shipping cost affects satisfaction; product price does not. Product price showed a weak/unclear relationship (1-star orders had the highest average price, 164.91, suggesting higher-paying customers may be more demanding, but the pattern is inconsistent). Shipping cost showed a clear relationship: average freight drops from 28.16 (1 star) to 21.71 (5 stars). Higher shipping costs are linked to worse reviews, likely tied to distant regions that also take longer to deliver.

Finding 3 — Product category affects satisfaction. Best-rated categories are simple, small, easy-to-ship items: books (4.54), food & drink (4.45), stationery (4.30). Worst-rated are large or complex items with higher expectations: office furniture (3.65), audio (3.84), male clothing (3.82). High-volume low-rated categories (office_furniture; bed_bath_table with 9,191 orders at 4.01) represent real business problems, not small-sample noise. Only categories with 100+ orders were kept, for statistical reliability.

Pivot tables

Review score by delivery speed:

- Fast (0-7 days): 4.41 | Medium (8-15 days): 4.28 | Slow (16+ days): 3.57

The score collapses once delivery passes ~15 days — a key actionable threshold for the business.

Cross pivot (delivery speed x freight level): delivery speed is the dominant factor. Slow deliveries score worst (3.36-3.62) regardless of shipping cost; fast deliveries score best (4.10-4.46). A slow delivery with cheap shipping (3.36) is worse than a fast delivery with expensive shipping (4.10) — customers value speed over low shipping cost.

Share (Visualizations)

Built a Tableau dashboard with four visualizations: (1) average review score by delivery speed, (2) top 10 highest-rated product categories, (3) bottom 10 lowest-rated categories, and (4) a heatmap crossing delivery speed with shipping cost.

Link: [View the interactive dashboard on Tableau Public](#)

Act (Conclusions & Recommendation)

- 1. Delivery time is the strongest driver of satisfaction.** Average delivery drops from 20.8 days (1 star) to 10.2 days (5 stars), and scores collapse past ~15 days.
- 2. Beating the delivery estimate matters.** 5-star orders arrived ~13 days early vs ~4 days for 1-star orders.
- 3. Shipping cost affects reviews; product price does not.**
- 4. Product category matters.** Simple items (books, food) rate highest; large or complex items (office furniture, audio) rate lowest.

Recommendation: Olist should prioritize delivery speed (keep under 15 days, beat estimates) and focus quality/logistics improvements on low-rated, high-volume categories like office furniture and bed_bath_table.